

OVERVIEW

The Main Board is the compute and interface core of Aura, a custom AI-assistant device. It packs 135 components onto a Ø73 mm round, 4-layer PCB shaped to sit behind a circular 480×480 display. Built around an STM32H743 Cortex-M7 running at 480 MHz, it drives the display, captures stereo audio, connects over Wi-Fi/BLE, and takes a single 5 V feed from the Power Board over an FPC, distributing it into cleanly separated supply rails.

AT A GLANCE

MCU STM32H743ZIT6 · Cortex-M7 @ 480 MHz	Stack-up 4-layer: Signal / GND / PWR / Signal
Form factor Ø73 mm round · 135 components	Power input 5 V from Power Board (FPC)
Rails 3V3_MCU · 3V3_WIFI · 3V3_MIC · 3V3_VDDIO · 5V_AUDIO	Tools Altium Designer · Blender

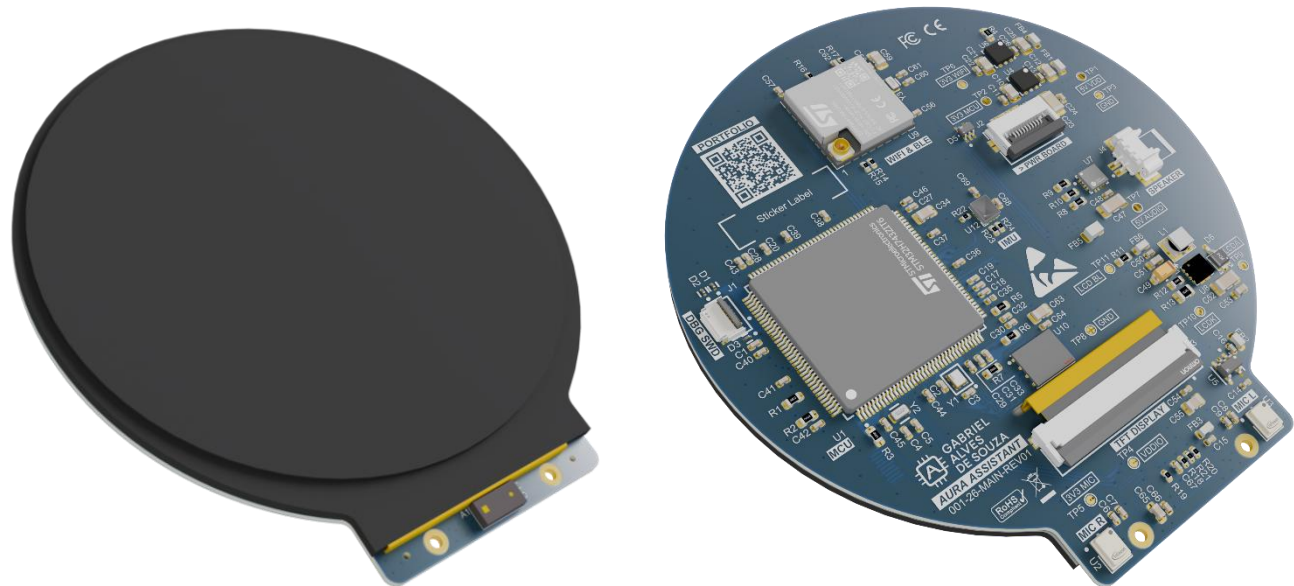
SYSTEM BLOCKS

- **Display** — 480×480 circular TFT, parallel RGB (LTDC)
- **Audio in** — dual MEMS microphone array (SAI / I²S)
- **Audio out** — Class-D speaker driver (MAX98357A)
- **Connectivity** — Wi-Fi / BLE module (ST67W611M1)
- **Memory** — external QSPI NOR Flash (assets & firmware)
- **Sensing** — IMU + time-of-flight gesture
- **Debug / USB** — SWD header + USB
- **Power** — STM32H7 distribution, per-rail regulation

ENGINEERING HIGHLIGHTS

- **Signals grouped by role** with 50 Ω controlled impedance and per-group length matching (QSPI 133 MHz, Wi-Fi SPI 100 MHz, LTDC RGB, SAI audio).
- **Power integrity** verified by current-density simulation on the 5V_VDD net; voltage drop well inside a 3% budget on a solid, uninterrupted plane.
- **EMI managed at the source:** stitching capacitors at reference-plane transitions, and isolation of the noisy switching blocks (Class-D amp, backlight boost).
- **Aggressor / victim net strategy:** sensitive audio routed clear of the fast buses to protect signal-to-noise.

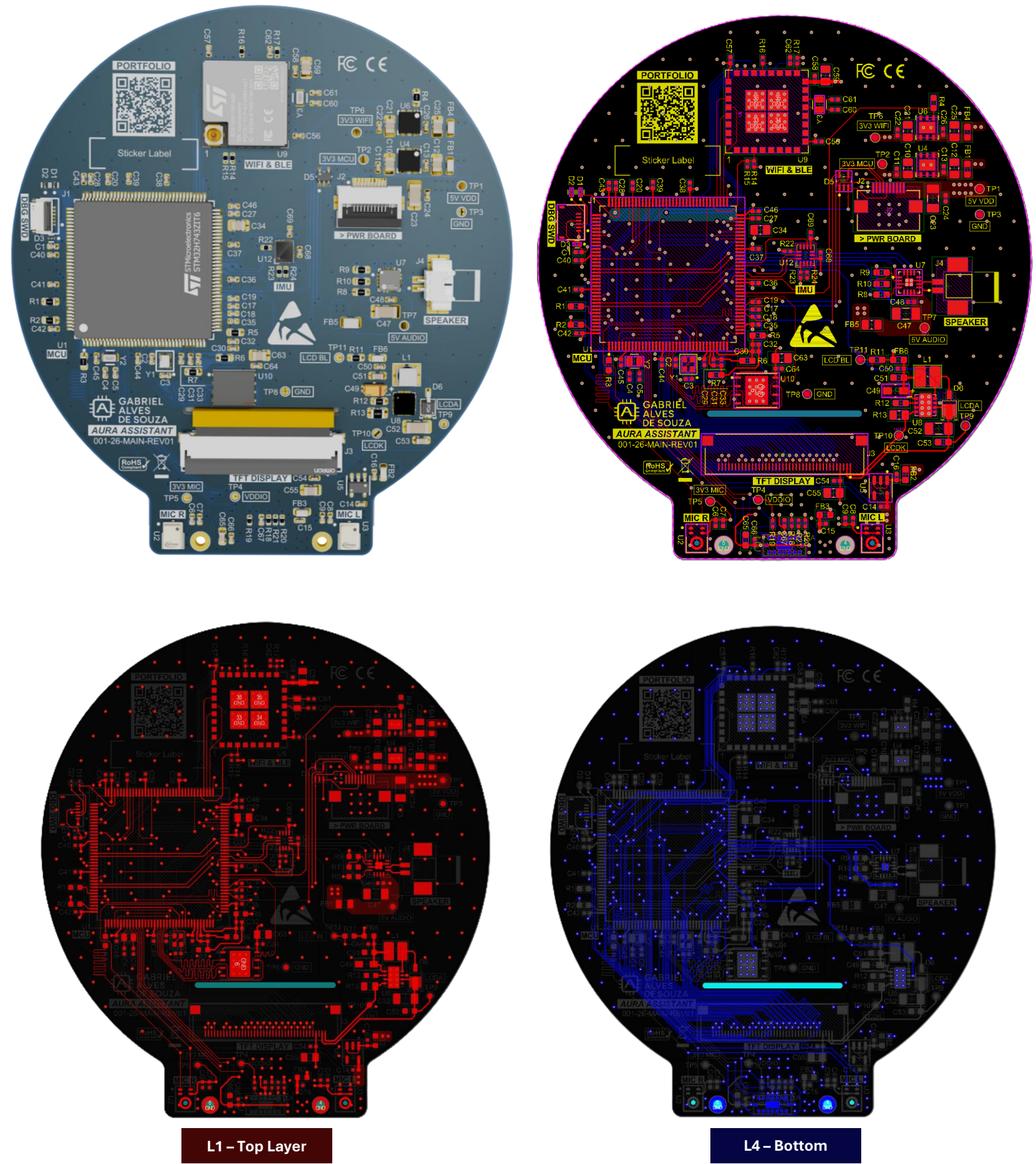
PCB 3D View



Aura Assistant - Main Board

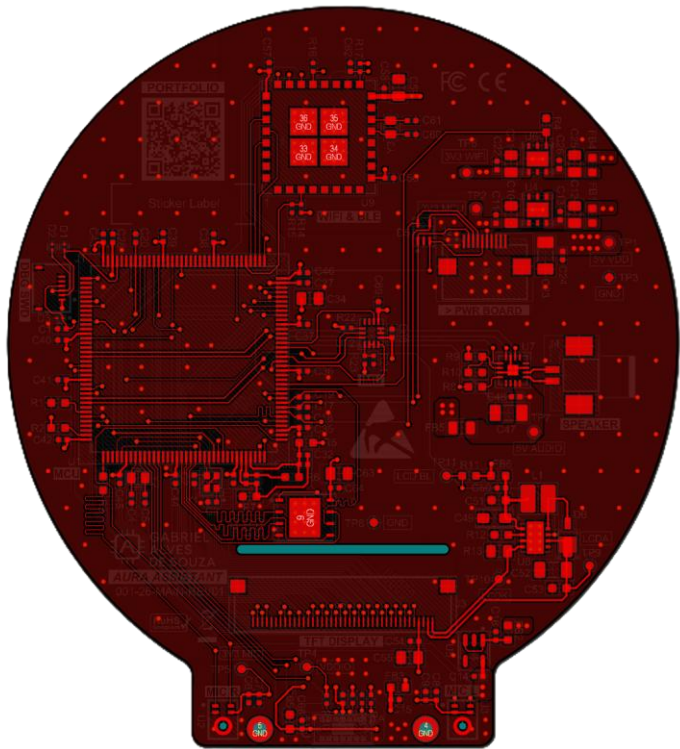
PCB_001-26-AURA_MAIN-REV01 · 4-layer · Ø73 mm round · 135 components

Altium Designer PCB Screenshots

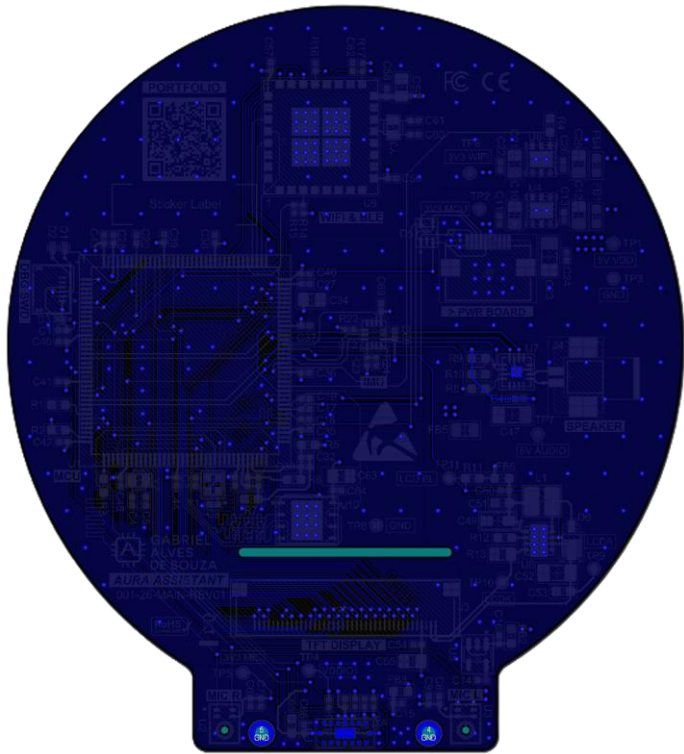


Altium Designer PCB Screenshots

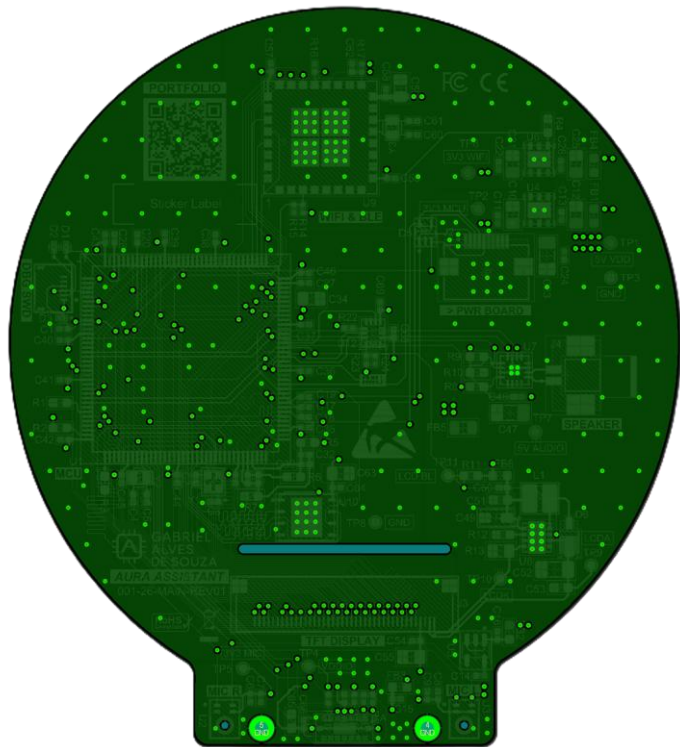
L1 – Top Layer



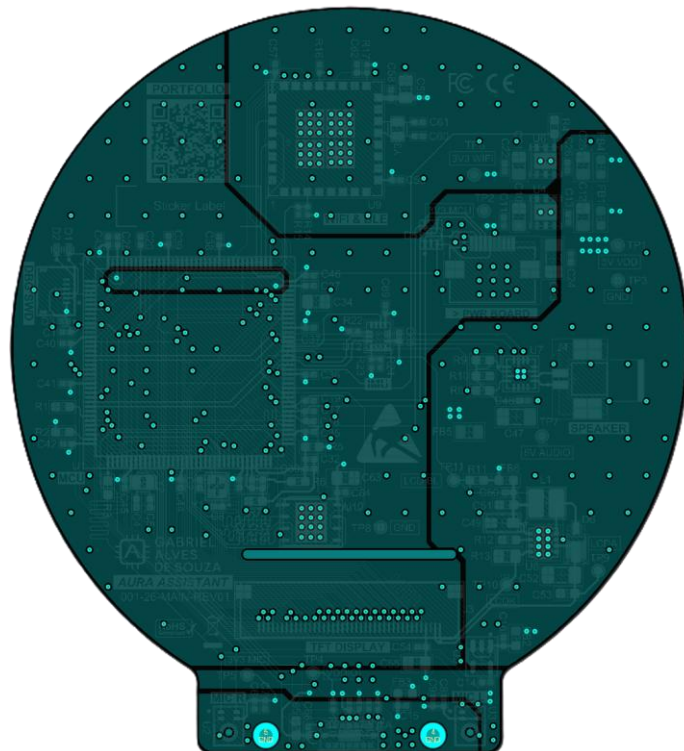
L4 – Bottom



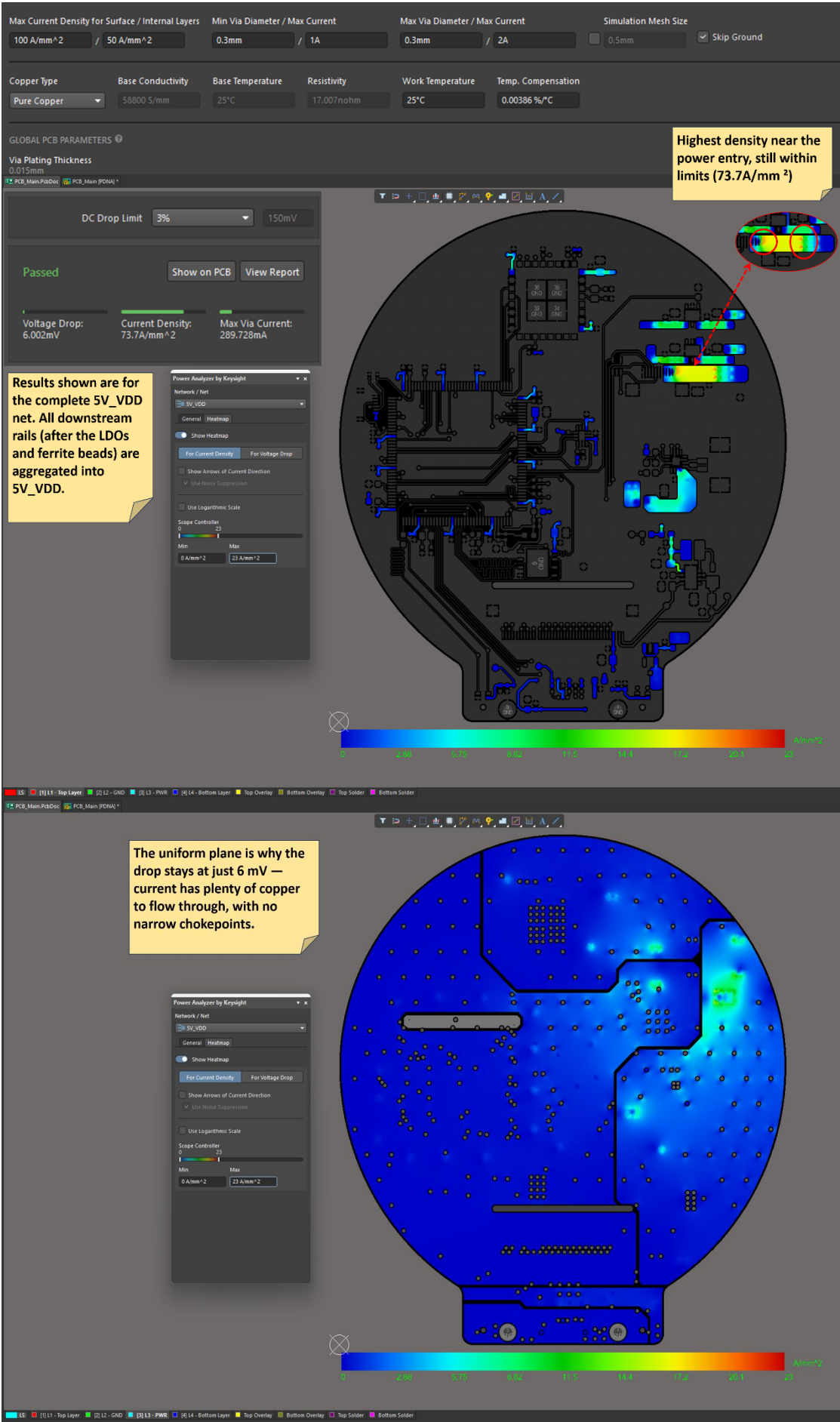
L2 – GND



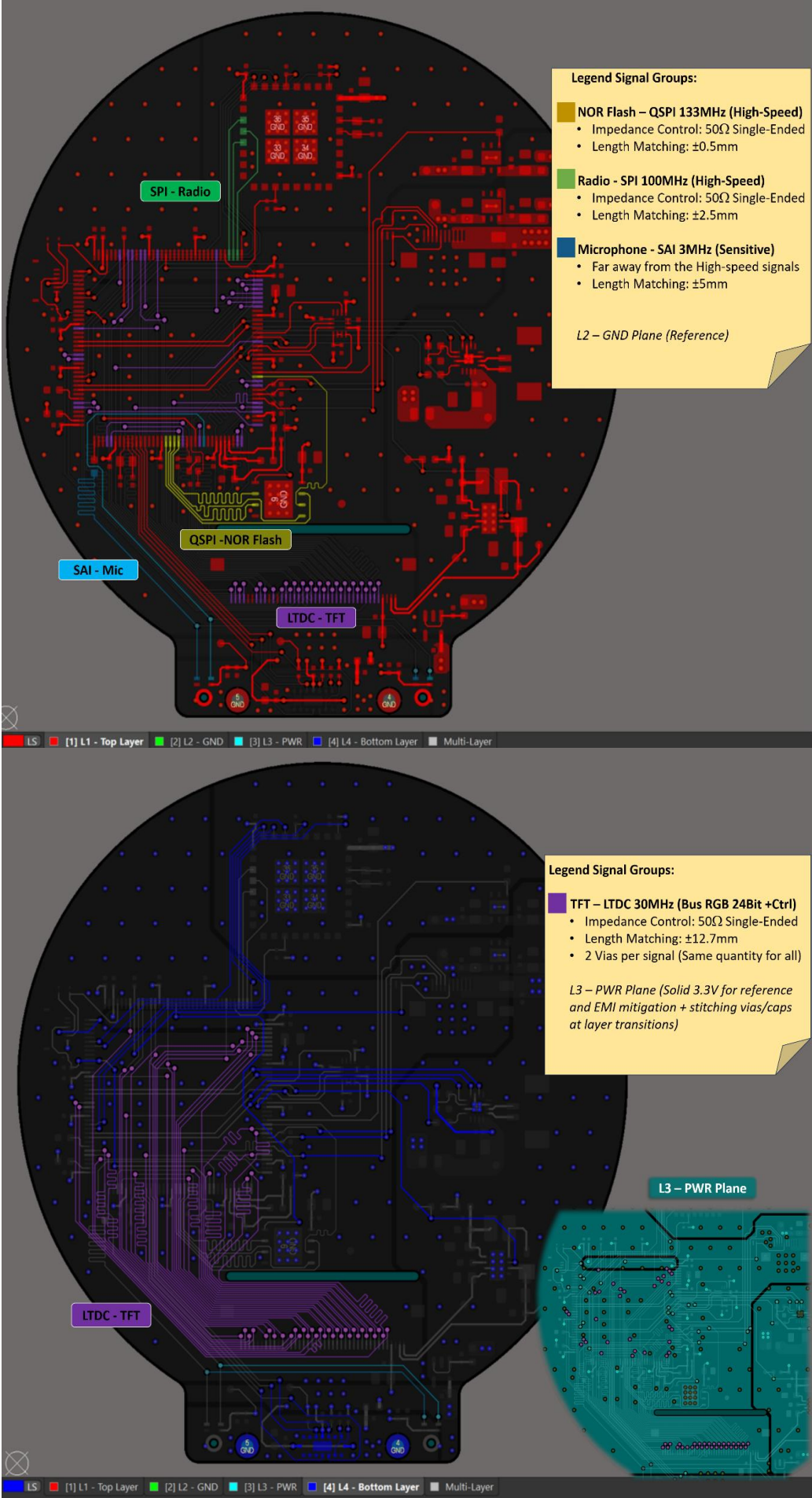
L3 – PWR



Power Integrity Analysis



Signal Groups & Routing Constraints



Signal Integrity Analysis

QUADSPI

Specifications: SingleEnded_50_Ohm_Impedance

Constraints Check Summary

✓ Impedance : 45-55Ohm

This parameter complies with the specified constraints

✓ Delay : <500ps

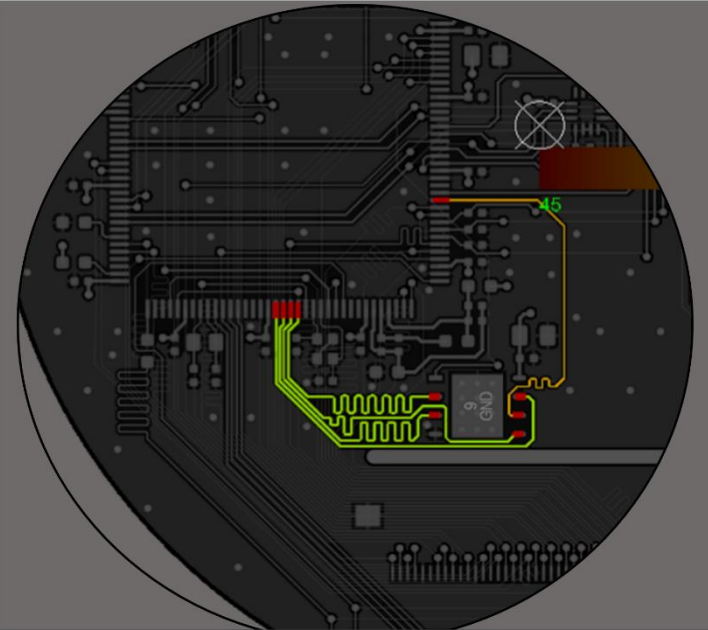
This parameter complies with the specified constraints

✓ Insertion Losses (IL) : <-20dB, 2GHz

This parameter complies with the specified constraints

✓ Return Losses (RL) : <-15dB, 2GHz

This parameter complies with the specified constraints



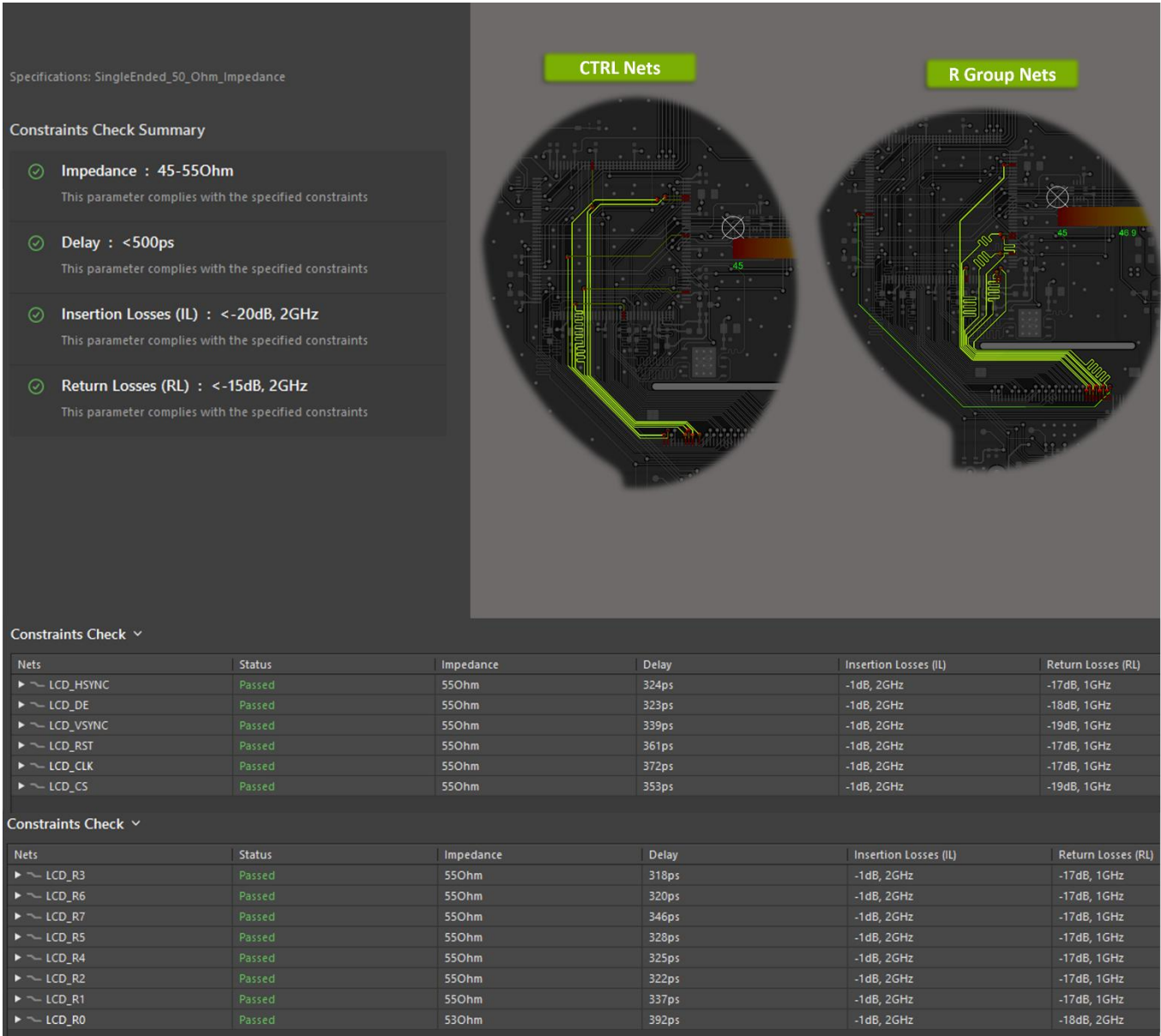
Layer Stackup ▾

#	Layer Name	Material	Type	Thickness
1	Top Overlay		Overlay	
2	Top Solder	SM-001	Solder Mask	0.0152mm
3	L1 - Top Layer		Signal	0.035mm
4	Dielectric 2	3313	Prepreg	0.0994mm
5	L2 - GND		Signal	0.0152mm
6	Dielectric 1		Core	0.5mm
7	L3 - PWR		Signal	0.0152mm
8	Dielectric 3	3313	Prepreg	0.0994mm
9	L4 - Bottom Layer		Signal	0.035mm
10	Bottom Solder	SM-001	Solder Mask	0.0152mm
11	Bottom Overlay		Overlay	

Constraints Check ▾

Nets	Status	Impedance	Delay	Insertion Losses (IL)	Return Losses (RL)
▶ ~ QUADSPI_IO3	Passed	48Ohm	171ps	-0dB, 2GHz	-28dB, 2GHz
▶ ~ QUADSPI_IO2	Passed	48Ohm	169ps	-0dB, 2GHz	-28dB, 2GHz
▶ ~ QUADSPI_IO1	Passed	48Ohm	169ps	-0dB, 2GHz	-28dB, 2GHz
▶ ~ QUADSPI_IO0	Passed	48Ohm	168ps	-0dB, 2GHz	-29dB, 2GHz
▶ ~ QUADSPI_CLK	Passed	53Ohm	167ps	-0dB, 2GHz	-19dB, 2GHz

Signal Integrity Analysis



Signal Integrity Analysis

